

Secure Banking System

on Amazon Web Services

A hands-on cloud security build: VPC networking, least-privilege Security Groups, EC2 web server, and an isolated RDS MySQL database — documented end-to-end from the AWS Console and Linux terminal.

What This Project Demonstrates

1 Network Isolation

Custom VPC with a dedicated public subnet, purpose-built for the application tier.

2 Least-Privilege Security

Two scoped Security Groups — EC2-SG and RDS-SG — so the database only ever hears from the app server.

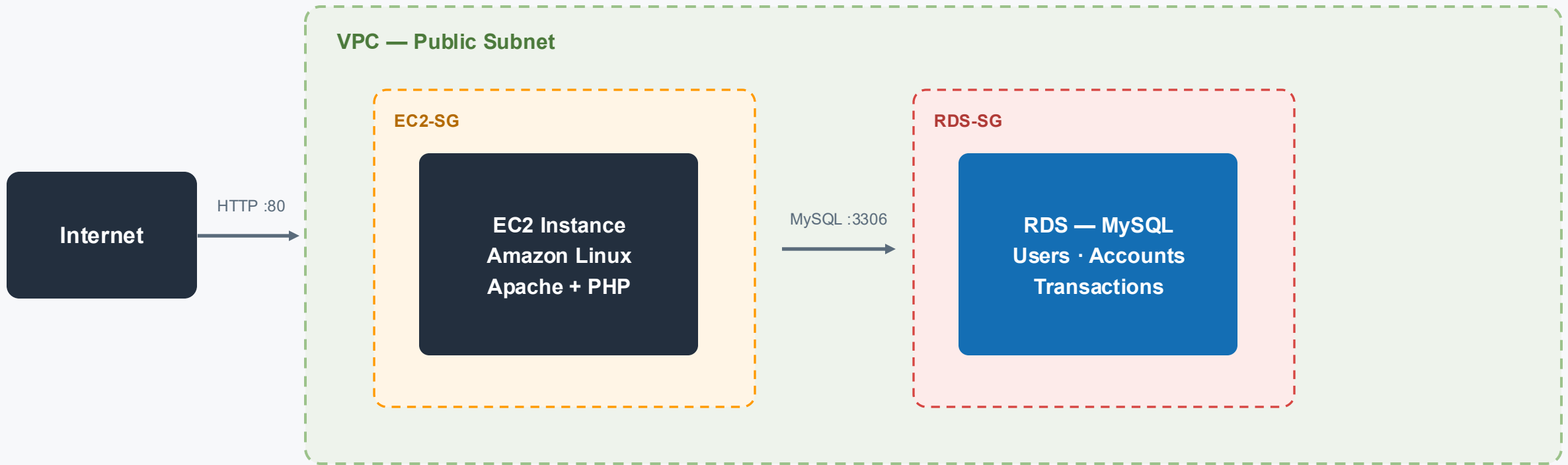
3 Live Web Application

Amazon Linux EC2 instance running Apache + PHP, serving a working banking demo.

4 Managed Relational Database

Amazon RDS (MySQL) storing Users, Accounts, and Transactions with proper foreign keys.

Architecture at a Glance



No public access to the database — RDS-SG accepts MySQL traffic only from EC2-SG.

Building the VPC

VPC dashboard

Filter by VPC

AWS Global View

Virtual private cloud

- Your VPCs
- Subnets
- Route tables
- Internet gateways
- Egress-only internet gateways
- Carrier gateways
- DHCP option sets
- Elastic IPs
- Managed prefix lists
- NAT gateways
- Peering connections
- Route servers

Security

Your VPCs (1/3) Info

Find VPCs by attribute or tag

Name	VPC ID	State	Encryption c...	Encryption control ...	Block Public...	IP
☒ vpc	vpc-0dbbe41f02c4cc76e	Available	-	-	Off	17
☐ Work VPC	vpc-094e5774bf98716b6	Available	-	-	Off	10
☐ Lab VPC	vpc-07faa0091edb5de8d	Available	-	-	Off	10

vpc-0dbbe41f02c4cc76e / vpc

Details Resource map CIDRs Flow logs Tags Related resources Integrations

Details

VPC ID vpc-0dbbe41f02c4cc76e	State Available	Block Public Access Off	DNS hostnames Enabled
DNS resolution Enabled	Tenancy default	DHCP option set dopt-05074d8b60836e1b4	Main route table rtb-058e493b7d5d5e787
Main network ACL acl-0b6ff5f4b37b588	Default VPC Yes	IPv4 CIDR 173.33.0.0/16	IPv6 pool

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Created the Virtual Private Cloud that hosts every resource in the project — the network boundary the whole architecture lives inside.

Configuring the Subnet

The screenshot displays the AWS Management Console interface for the 'Subnets' page. The left sidebar shows the 'VPC dashboard' with a 'Subnets' link highlighted. The main content area shows a list of subnets under the heading 'Subnets (1/8)'. The selected subnet, 'subnet-0801ebb3ee33e83fa', is highlighted in blue. Below the list, the 'Details' tab for this subnet is expanded, showing various attributes.

Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR
-	subnet-0c9711d31345d4206	Available	vpc-0dbbe41f02c4cc76e vpc	Off	172.31.16.0/
-	subnet-06833579a3d15a93c	Available	vpc-0dbbe41f02c4cc76e vpc	Off	172.31.64.0/
-	subnet-01ddb4da19bfefe8c	Available	vpc-0dbbe41f02c4cc76e vpc	Off	172.31.32.0/
-	subnet-0801ebb3ee33e83fa	Available	vpc-0dbbe41f02c4cc76e vpc	Off	172.31.80.0/

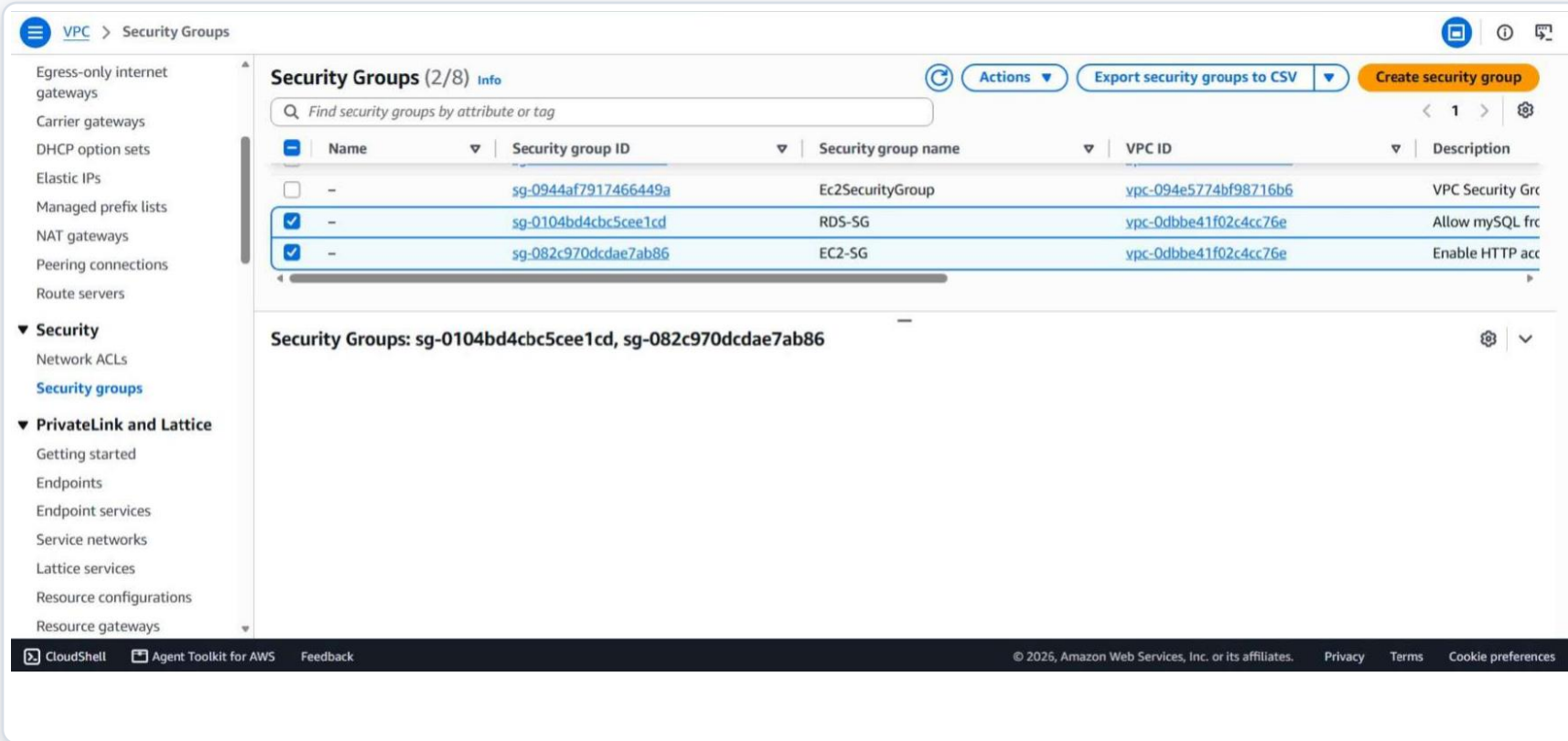
subnet-0801ebb3ee33e83fa

Details

Subnet ID subnet-0801ebb3ee33e83fa	Subnet ARN arn:aws:ec2:us-east-1:676261197495:subnet/subnet-0801ebb3ee33e83fa	State Available	Block Public Access Off
IPv4 CIDR 172.31.80.0/20	Available IPv4 addresses 4090	IPv6 CIDR -	IPv6 CIDR association ID -
Availability Zone us-east-1b	Network border group	VPC vpc-0dbbe41f02c4cc76e vpc	Route table rtb-058e493b7d5d5e787

Selected a public subnet with auto-assigned public IPv4, so the web server can be reached from the internet while everything else stays private.

Least-Privilege Security Groups



The screenshot shows the AWS Management Console for Security Groups. The left sidebar lists various VPC resources, with 'Security groups' highlighted under the 'Security' section. The main panel displays a table of security groups. Two custom groups are selected: RDS-SG and EC2-SG. Below the table, a summary shows the selected security groups: sg-0104bd4cbc5cee1cd and sg-082c970dcdae7ab86.

	Name	Security group ID	Security group name	VPC ID	Description
☐	-	sg-0944af7917466449a	Ec2SecurityGroup	vpc-094e5774bf98716b6	VPC Security Gr...
☒	-	sg-0104bd4cbc5cee1cd	RDS-SG	vpc-0dbbe41f02c4cc76e	Allow mySQL fr...
☒	-	sg-082c970dcdae7ab86	EC2-SG	vpc-0dbbe41f02c4cc76e	Enable HTTP acc...

Security Groups: sg-0104bd4cbc5cee1cd, sg-082c970dcdae7ab86

Two custom groups: EC2-SG opens HTTP to the internet. RDS-SG allows MySQL (3306) only from EC2-SG — the database is invisible to the public internet.

Launching the Web Server

The screenshot displays the AWS Management Console's EC2 Instances page. On the left, a navigation sidebar lists various AWS services, with 'Instances' selected. The main panel shows a table of three EC2 instances. The first instance, 'Banking-App-Server', is selected and highlighted. Below the table, the details for this instance are expanded, showing its ID, type, state, and network configuration.

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS
Banking-App-Server	i-0cc5e7397177bfa12	Running	t2.micro	2/2 checks passed	us-east-1b	ec2-107-23-6-218.com...
Lab Instance	i-0c663b61dc8a4513a	Running	t3.medium	3/3 checks passed	us-east-1a	ec2-52-2-39-69.comput...
Bastion Host	i-0e95811b8b77d85df	Running	t2.micro	2/2 checks passed	us-east-1a	ec2-100-53-225-197.co...

i-0cc5e7397177bfa12 (Banking-App-Server)

Instance summary

- Instance ID: i-0cc5e7397177bfa12
- Public IPv4 address: 172.31.95.181 (open address)
- Private IPv4 addresses: 172.31.95.181
- Instance state: Running
- Public DNS: ec2-107-23-6-218.compute-1.amazonaws.com (open address)

Banking-App-Server: a t2.micro Amazon Linux instance, Running with 2/2 status checks passed and a public IPv4 address ready for traffic.

Apache + PHP — First Boot

Banking App Server is Running!

PHP Version: 8.5.9

Apache and PHP were installed automatically via an EC2 user-data script on first boot — the server was serving pages before any manual login.

Connecting EC2 to RDS

```

Running scriptlet: mariadb105-3:10.5.29-1.amzn2023.0.1.x86_64 5/5
Verifying      : mariadb-connector-c-3.3.10-1.amzn2023.0.2.x86_64 1/5
Verifying      : mariadb-connector-c-config-3.3.10-1.amzn2023.0.2.noarch 2/5
Verifying      : mariadb105-3:10.5.29-1.amzn2023.0.1.x86_64 3/5
Verifying      : mariadb105-common-3:10.5.29-1.amzn2023.0.1.x86_64 4/5
Verifying      : perl-Sys-Hostname-1.23-477.amzn2023.0.9.x86_64 5/5

Installed:
  mariadb-connector-c-3.3.10-1.amzn2023.0.2.x86_64      mariadb-connector-c-config-3.3.10-1.amzn2023.0.2.noarch      mariadb105-3:10.5.29-1.amzn2023.0.1.x86_64
  mariadb105-common-3:10.5.29-1.amzn2023.0.1.x86_64      perl-Sys-Hostname-1.23-477.amzn2023.0.9.x86_64

Complete!
[ec2-user@ip-172-31-95-181 ~]$ mysql -h database-██████████.us-east-1.rds.amazonaws.com -u admin -p
Enter password:
ERROR 1045 (28000): Access denied for user 'admin'@'172.31.95.181' (using password: YES)
[ec2-user@ip-172-31-95-181 ~]$ mysql -h database-██████████.us-east-1.rds.amazonaws.com -u admin -p
Enter password:
ERROR 1045 (28000): Access denied for user 'admin'@'172.31.95.181' (using password: YES)
[ec2-user@ip-172-31-95-181 ~]$ mysql -h database-██████████.us-east-1.rds.amazonaws.com -u admin -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MySQL connection id is 42
Server version: 8.4.9 Source distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MySQL [(none)]>

```

i-0cc5e7397177bfa12 (Banking-App-Server)

PublicIPs: 107.23.6.218 PrivateIPs: 172.31.95.181

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From the EC2 terminal, installed a MySQL client and connected to the RDS endpoint. Access was denied until the correct credentials confirmed the security boundary was working — then the connection succeeded.

Designing the Database Schema

```
-> account_id INT AUTO_INCREMENT PRIMARY KEY,  
-> user_id INT,  
-> balance DECIMAL(10,2) DEFAULT 0,  
-> FOREIGN KEY (user_id) REFERENCES Users(user_id)  
-> );  
Query OK, 0 rows affected (0.108 sec)  
  
MySQL [bankingdb]> CREATE TABLE Transactions (  
-> transaction_id INT AUTO_INCREMENT PRIMARY KEY,  
-> sender INT,  
-> receiver INT,  
-> amount DECIMAL(10,2),  
-> date DATETIME DEFAULT CURRENT_TIMESTAMP,  
-> FOREIGN KEY (sender) REFERENCES Accounts(account_id),  
-> FOREIGN KEY (receiver) REFERENCES Accounts(account_id)  
-> );  
Query OK, 0 rows affected (0.044 sec)  
  
MySQL [bankingdb]> SHOW TABLES;  
+-----+  
| Tables_in_bankingdb |  
+-----+  
| Accounts              |  
| Transactions          |  
| Users                 |  
+-----+  
3 rows in set (0.009 sec)  
  
MySQL [bankingdb]>
```

i-0cc5e7397177bfa12 (Banking-App-Server)

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Created the Accounts and Transactions tables with foreign keys back to Users and Accounts — SHOW TABLES confirms all three tables exist inside bankingdb.

Seeding & Verifying Test Data

```
| Accounts      |
| Transactions  |
| Users         |
+-----+
3 rows in set (0.009 sec)

MySQL [bankingdb]> INSERT INTO Users (name, email) VALUES ('Ahmed Test', 'ahmed@test.com');
Query OK, 1 row affected (0.010 sec)

MySQL [bankingdb]> INSERT INTO Accounts (user_id, balance) VALUES (1, 1000.00);
Query OK, 1 row affected (0.013 sec)

MySQL [bankingdb]> SELECT * FROM Users;
+-----+
| user_id | name      | email      |
+-----+
| 1       | Ahmed Test | ahmed@test.com |
+-----+
1 row in set (0.001 sec)

MySQL [bankingdb]> SELECT * FROM Accounts;
+-----+
| account_id | user_id | balance |
+-----+
| 1          | 1       | 1000.00 |
+-----+
1 row in set (0.001 sec)

MySQL [bankingdb]>
```

i-0cc5e7397177bfa12 (Banking-App-Server)

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Inserted a sample user and linked account, then queried both tables directly in MySQL to confirm the relational data was stored correctly before wiring up the app.

PHP — Connecting to RDS

```

GNU nano 8.3 /var/www/html/index.php Modified
<?php
$servername = "database-banking.cprw6iihuaro.us-east-1.rds.amazonaws.com";
$username = "admin";
$password = "XXXXXXXXXX";
$dbname = "bankingdb";

$conn = new mysqli($servername, $username, $password, $dbname);

if ($conn->connect_error) {
    die("Connection failed: " . $conn->connect_error);
}

echo "<h1>Banking App - Connected to Database</h1>";

$sql = "SELECT u.name, u.email, a.balance FROM Users u JOIN Accounts a ON u.user_id = a.user_id";
$result = $conn->query($sql);

if ($result->num_rows > 0) {
    echo "<table border='1' cellpadding='10'>";
    echo "<tr><th>Name</th><th>Email</th><th>Balance</th></tr>";
    while($row = $result->fetch_assoc()) {
        echo "<tr><td>" . $row["name"]. "</td><td>" . $row["email"]. "</td><td>" . $row["balance"]. "</td></tr>";
    }
    echo "</table>";
} else {

^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo      M-A Set Mark  M-] To Bracket M-E Previous
^X Exit      ^R Read File  ^N Replace    ^U Paste      ^J Justify    ^/_ Go To Line M-B Redo      M-G Copy      ^E Where Was  M-R Next

i-0cc5e7397177bfa12 (Banking-App-Server)
PublicIPs: 107.23.6.218 PrivateIPs: 172.31.95.181

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```

index.php opens a mysqli connection to the RDS endpoint and runs a JOIN across Users and Accounts, rendering the result as an HTML table.

Live — Data Rendered From RDS

Banking App - Connected to Database

Name	Email	Balance
Ahmed Test	ahmed@test.com	\$1000.00

Opening the EC2 public IP in a browser returns a live table — Name, Email, and Balance — pulled straight from the RDS MySQL database.

End-to-end proof: Internet → EC2 (Apache/PHP) → RDS (MySQL) → back to the browser.

Skills Demonstrated

● AWS VPC & Subnetting

● Security Groups (Least Privilege)

● Amazon EC2 (Amazon Linux)

● Amazon RDS (MySQL)

● Linux Command Line

● Apache & PHP

● Relational Schema Design

● EC2 User-Data Automation